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James H. Simons, PhD:
Using Mathematics to Make Money



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USING MATHEMATICS TO MAKE MONEY

James H. Simons, PhD, is chairman of the Simons Foundation, an organization dedicated to advancing the frontiers of research in mathematics and the basic sciences. The foundation's philanthropic activities include a major research initiative on the causes of autism and establishment of an institute for research in mathematics and theoretical physics. Particularly focused on the growing interface between the physical and life sciences, the foundation has established and endowed several such research programs at universities and institutions both in the United States and abroad.

Simons founded Renaissance Technologies LLC, a highly quantitative investment firm, from which he retired in 2009, after serving as its chief executive officer for more than thirty years. Previously, he was chair of the Mathematics Department at the State University of New York at Stony Brook. Earlier in his career, he was a cryptanalyst at the Institute of Defense Analyses (then located in Princeton, New Jersey), and taught mathematics at the Massachusetts Institute of Technology and Harvard University.

Simons is the founder and chair of Math for America, a nonprofit organization whose mission is to improve math education in U.S. public schools. He serves as a trustee of Brookhaven National Laboratory, the Institute for Advanced Study, Rockefeller University, the New York Genome Center, Institut des Hautes Études Scientifiques, and the Mathematical Sciences Research Institute in Berkeley. In addition, he is a member of the board of the MIT Corporation and chair emeritus of the Stony Brook Foundation, as well as a member of the American Academy of Arts and Sciences, the American Philosophical Society, and the National Academy of Sciences.

Simons earned a BS in mathematics from the Massachusetts Institute of Technology and a PhD in mathematics from the University of California, Berkeley. His scientific research was in the area of geometry and topology. He received the American Mathematical Society's Oswald Veblen Prize in Geometry in 1976 for work that involved recasting the subject of area-minimizing multi-dimensional surfaces. Simons' most influential research involved the discovery and application of



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certain geometric measurements, now called the Chern-Simons invariants, which have wide use, particularly in theoretical physics.

In September 2022, James Simons spoke with members of the Journal of Investment Consulting editorial board about how his experience as a mathematician prepared him for success in the financial world; the main ingredients of his quantitative investment firm's success—building and continuously improving investment models through regular testing; hiring excellent scientists who have decided they want to make money rather than people with previous experience in finance; and fostering collaboration among the company's workforce. Other topics covered in the discussion included current competition in the quantitative approach to investing, the difficulty of predicting how advances in fundamental research will eventually be applied in the real world, and the numerous philanthropic activities Simons supports. Taking part in the discussion were Inna Okounkova, Columbia University and editor-in-chief of the Journal; Edward Baker, Mesirow Financial; Ludwig Chincarini, University of San Francisco and United States Commodity Funds; Arun Muralidhar, Mcube Investment Technologies; and Margaret Towle, Yakima River Partners.

Edward Baker: You were an extremely successful mathematician in the early part of your career with some path-breaking work in differential geometry. And your work with Shiing-Shen Chern in particular was important to the future of that field and many fields in mathematics.¹ Then you went into finance. Has your deep training in mathematics helped you? Or has that just been interesting but superfluous to your success in finance?

James Simons: Well, as you know, Renaissance is very quantitative, and the fact that I was a mathematician was somewhat helpful. It gave me the ability to tell who was good and who wasn't.

Edward Baker: In terms of your employees, you mean?

James Simons: Yes. The work with Chern was amazing, because what's called Chern-Simons theory is now embedded in physics.² Every day an average of four papers come out in the physics journals, or in the archive, referencing and using

Chern-Simons. It's particularly astounding because neither of us knew any physics. Well, maybe Chern knew a little, but I didn't know any physics.

Edward Baker: It's typical of mathematics, isn't it, that when new ideas come to light, they don't seem to have anything to do with the applied world. And then in many cases, they become hugely important.

James Simons: Well, I certainly like mathematics, and I was a pretty good mathematician. Then I started trying to make money.

Edward Baker: And has the mathematics been useful, or has it just helped you identify talent?

James Simons: I understood that people would come up with new ideas that I was able to understand because I am a mathematician. But mostly, hiring great people was beneficial.

Edward Baker: Has collaboration been as important in your research team as it is in mathematics?

James Simons: At Renaissance, collaboration has been extremely important.

Margaret Towle: Over the course of your career, Jim, you've given a number of interviews. In one of them in 2020, you talked about the secret sauce of what made you successful. You said the first ingredient was hiring great people, and the second one was providing great infrastructure. What you did early on was to create an infrastructure that was not just out of the box—it was a brand-new box. And you were fostering collaboration as Ed mentioned, and sharing profits with everyone.

In your family office and especially in your private foundation, there's a different model in terms of sharing the profits. How have you compensated for this missing element? How do you run the private foundation or your nonprofit activities to take advantage of what I consider an important tenet of successful organizations?

James Simons: Obviously we're a not-for-profit, so we don't have any profits to share. We hire scientists in various areas. Math and physical science constitute one area in which we offer grants. We also offer grants on the biological side. We also created the Flatiron Institute,³ which now has five groups of about fifty people each: computational biology, computational astrophysics, computational quantum physics, computational mathematics, and computational neuroscience.

These groups don't have to apply for grants, like many scientists have to, because we pay them. I won't say these people are not interested in making money, but they get good salaries,

and the best people get the highest salaries, more or less. The people are terrific, and they've accomplished an awful lot. The Flatiron Institute is called that because it's in the Flatiron District in New York City. I'm very proud of it.

Margaret Towle: It sounds like you've figured out a way to motivate group members relative to their interests. Applying for grants is time-consuming and often involves a lot of administrative hassle. So perhaps you've substituted salaries for a share of the profits.

James Simons: Well, if you think a share of the profits is not having to apply for a grant.

Inna Okounkova: On the topic of philanthropy, Jim, you and your wife Marilyn have provided substantial financial support to various causes in many fields. In the field of mathematics specifically, your funding is probably comparable to the support of the National Science Foundation, and it touches all levels of mathematical education from middle school to universities to scientific institutes. The next International Congress of Mathematics will be in Philadelphia, thanks to your generosity, and we're all grateful to you for that. But how satisfied are you with the impact you've made on mathematical research and education so far?

James Simons: Let's talk about education. We started a non-profit called Math for America, which, in its early years, focused on recognizing outstanding teachers in New York City—identifying the best teachers and bringing them into the fold.⁴ They had a four-year run, and they received \$15,000 a year on top of their regular salaries. After four years, if they had done a really good job, they could apply for another four years. At a certain point, we decided to include science as well as mathematics. Currently, about half of the high school courses cover mathematics, and the other half deal with biology, chemistry, and physics. It's been a remarkable project. It's been going on for maybe twenty years, and it costs us \$25 million a year to run. Now, it's going to be national.

In the science bills that just passed Congress, there's a provision that's going to take nationwide what we've been doing in New York City.⁵ I tried to get that done for years. We approached Obama with the idea; he liked it, but his Department of Education head only wanted to fix bad teachers. We're doing the opposite. We're identifying and rewarding outstanding teachers. So now this program will be nationwide. It's going to take a while to open it up across the country because the program needs to be approved by the Appropriations Committee, so we hope that won't be a stumbling block.

Ludwig Chincarini: A number of portfolio managers who have left Renaissance's Medallion Fund—I'm thinking in particular of Sandor Straus⁶—have not been able to replicate the

performance of Medallion. What is the secret of the success of the Medallion Fund, in terms of broad investment strategies? If you want, please touch on matters like whether you have automatic trading or traders, what types of data and price predictors you rely on, how you save on transaction costs, and what types of assets you invest in globally—futures, equities?

James Simons: The secret of Medallion's success is hard work over a long period of time by a team of world-class scientists who take a mathematical approach to investing.

Ludwig Chincarini: Who leads that team of world-class scientists now?

James Simons: Peter Brown has served as Co-CEO and then as sole CEO since I left Renaissance 13 years ago.

Ludwig Chincarini: My original question was simply what is the trick or the secret sauce that makes Medallion successful? But when you said it's hard to replicate, I also asked why it's so hard to replicate.

James Simons: It's hard to replicate because it took thousands of person-years to build it.

Ludwig Chincarini: Do most of Medallion's model-building math and statistics basically rely on predictions from those tools? Or do you also consider other popular trends, such as behavioral bias, in which some funds and some people believe humans make mistakes and investors can take advantage of those mistakes? Or do you have an agnostic view regarding those types of considerations and simply stick to the math and the predictions?

James Simons: We stick to the math.

Arun Muralidhar: The big issue in our business is separating signal from noise or skill versus luck. Would you elaborate on how math helped you as you went about identifying good strategies or separating the signal from the noise?

James Simons: The mathematics I did was essentially training my mind; none of it had anything to do with making money. Although Chern-Simons was wonderful, it does not help at all with making money.

Edward Baker: What sort of longevity do you expect from your model?

James Simons: We are constantly changing the model to incorporate new information. Some aspects of it last for a very long time, others not as long, and some are just wrong from the beginning because we've mistaken noise for signal.

Edward Baker: Market cycles change, things don't work for a while, and then they'll work again. How do you deal with that?

James Simons: We have only one model, one system, which is improved continuously on the basis of new signals and market behavior in general.

Edward Baker: Do you sometimes change your model one way and then later find you have to change it back?

James Simons: Yes, that can happen.

Ludwig Chincarini: Correct me if I'm wrong, but is some of the testing you do a bit like data mining—meaning, are you testing lots of strategies until you find one that works well? And is that good or bad?

James Simons: Yes, we do a lot of data mining. We have thousands of computers scouring data for information related to the movement of tradeable instruments. Sometimes what we find works, and sometimes it does not.

Edward Baker: Do you think it's become harder to make money using quantitative approaches like this? There's certainly more competition, more people using this approach. Has that made it harder or not?

James Simons: I think this has made it harder. The more quantitative funds there are, the more competition we'll have. In business, you need to beat the competition. It's true that when we started, there were few quantitative funds. Maybe there were one or two. But you just have to try to stay ahead of the pack. We don't know what they're doing, but we know what we're doing, and we just have to get better and better at it.

Edward Baker: Obviously, one of the things that's changed since you started is the availability of massive datasets. Has that changed your approach?

James Simons: As new datasets become available, they are incorporated into what we're doing.

Arun Muralidhar: How has the field of artificial intelligence and machine learning influenced investment strategies in general, and how has it affected your own experience? When I talk to folks who operate funds in that space, one of the challenges they sometimes cite is their inability, in a machine learning system, to explain why they got the signal. The machine simply did it. What are your thoughts on this area of investments?

James Simons: Machine learning has come a long way, and is very important to our work at Renaissance and at Flatiron.